

Deterministic Optimization

Further Development of Simplex Method

Andy Sun

Assistant Professor

Stewart School of Industrial and Systems Engineering

Phase I/Phase II Simplex

Further Development of Simplex Method

Learning Objectives for this lesson

- Discover an algorithmic method to detect infeasibility of an LP and find a BFS when it exists

Initialization of Simplex Method

- Simplex method starts from a BFS
- First, there has to exist a BFS
- Second, a BFS needs to be given as input to the simplex method
- It is not easy to detect feasibility (i.e. checking if there exists a BFS) nor to find a BFS when it exists
- In fact, both problems are equivalent to solving an LP as we will see

A Little Philosophical Thinking

- How to detect feasibility of an LP?
 - Point out a feasible solution x
 - It is easy to verify if such a solution x is feasible by plugging x back into the linear system of LP
- How to detect infeasibility of an LP?
 - It is not useful if we just claim that we cannot find a feasible solution, because maybe we haven't tried hard enough
 - Prove that something does not exist can be quite hard (think about criminal justice, a suspect is first assumed innocent and needs to be proved guilty, not the other way around)
 - It turns out for LP, proving feasibility and infeasibility are equally easy

Idea of Two-Phase Simplex Method

- Since finding a BFS is not a trivial task, we device a two-phase simplex method:
 - The first phase solves a related LP, which is always feasible and is easy to write down a BFS so that the simplex method can be applied
 - The result of this phase I simplex method is an output indicating either the original LP is infeasible, therefore, termination of the entire algorithm; or a BFS of the original LP
 - In the second case, Phase II simplex method starts with the found BFS and solves the original LP

Phase I Simplex Method

- Original standard form LP:
 - (P) $\min c^\top x$
 - s.t. $Ax = b$
 - $x \geq 0$
 - Where $A \in R^{m \times n}$, $b \geq 0$ (we can always achieve this)

- Construct a new LP:
 - (P1) $z_{p1} = \min \sum_{i=1}^m y_i$
 - s.t. $Ax + y = b$
 - $x \geq 0, y \geq 0$

Claim:

- (P1) is always feasible
- If $z_{p1} = 0$, (P) is feasible and optimal solution x of (P1) is a BFS of (P)
- If $z_{p1} > 0$, (P) is infeasible

Why Is the Claim True?

- (P1) $z_{p1} = \min \sum_{i=1}^m y_i$
- s.t. $Ax + y = b$
- $x \geq 0, y \geq 0$
- First, $(x = 0, y = b)$ is a feasible solution of (P1), because it satisfies the equality constraint, and since $b \geq 0$, also satisfies nonnegativity
- Second, $(x = 0, y = b)$ is also a BFS of (P1), because the matrix corresponding to y is identity I , so we can start simplex on (P1) with this BFS
- Third, (P1) is always bounded below, because $y \geq 0, z_{p1} \geq 0$, so simplex method on (P1) always terminates with an optimal BFS

Why Is the Claim True? (cont.)

- (P1) $z_{p1} = \min \sum_{i=1}^m y_i$
- s.t. $Ax + y = b$
- $x \geq 0, y \geq 0$
- Fourth, if $z_{p1} = 0$, since $y \geq 0$, the optimal solution $y^* = 0$. So y^* is the nonbasic variable, and x^* is basic variable. Thus, x^* is also a BFS of (P)
- Fifth, if $z_{p1} > 0$, then (P) must be infeasible, because if (P) were feasible, there must be some x and $y = 0$ that is feasible to (P1) with objective value 0

Phase I/Phase II Simplex Method

1. Phase I:

Solve the Phase-I LP. Denote the optimal cost as z_{p1} . We will have two possibilities:

- (a) If $z_{p1} > 0$, the original LP is infeasible. The algorithm terminates.
- (b) If $z_{p1} = 0$, a feasible solution to the original LP is found, from which we can obtain a BFS for the original LP.

2. Phase II:

Solve the original LP by the simplex method, starting with the BFS found in Phase I.

Summary

- We learned:
 - How to detect feasibility/infeasibility of LP using Phase I LP
 - Phase I/Phase II Simplex Method